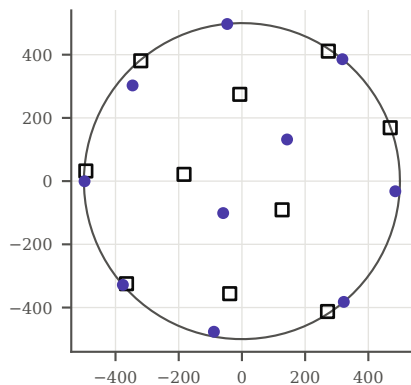


□ best layout of all methods ● best PSO-VNS layout

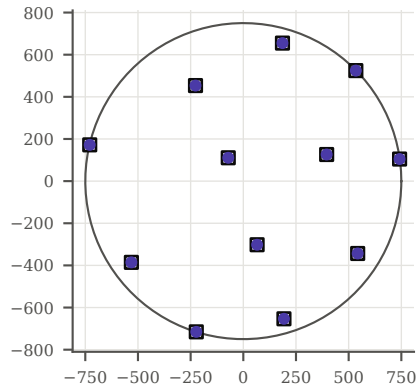
Data Set I, $r = 500$ m, $N = 10$

best of all methods: MS-SLSQP (best PSO-VNS: +0.21 pp loss)



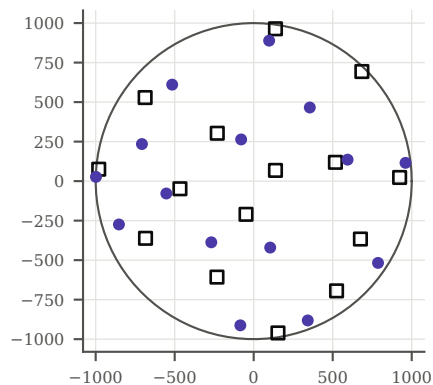
Data Set I, $r = 750$ m, $N = 12$

best of all methods: PSO-VNS

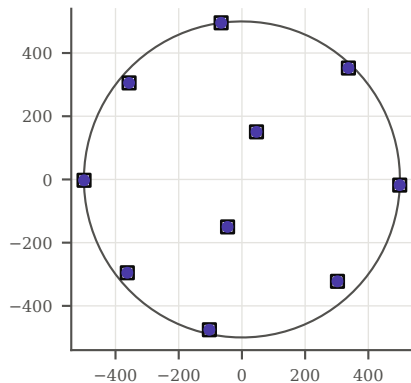


Data Set I, $r = 1000$ m, $N = 15$

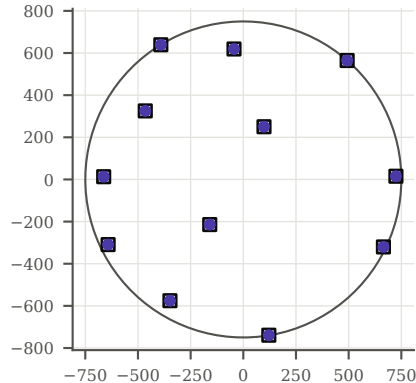
best of all methods: PSO (best PSO-VNS: +0.09 pp loss)



Data Set II, $r = 500$ m, $N = 10$
best of all methods: PSO-VNS



Data Set II, $r = 750$ m, $N = 12$
best of all methods: PSO-VNS



Data Set II, $r = 1000$ m, $N = 15$
best of all methods: PSO-VNS

